Transfer Learning for Parking Occupancy Detection: An Analysis with Limited Data

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***Abstract*—Automated parking space occupancy detection is critical for smart city infrastructure, reducing parking search time and urban congestion. We conduct a case study comparing transfer learning with ResNet18 fine-tuning against a custom CNN trained from scratch on limited parking occupancy data. We construct a dataset by extracting and cropping individual parking spaces from full parking lot scenes, then examine how pretrained ImageNet features adapt to parking-specific classification tasks. Through empirical analysis, we explore ResNet18's transfer learning performance, including feature initialization, regularization effects, and the impact of selective fine-tuning on small datasets. We characterize the relative performance of both approaches on our parking detection task and analyze the mechanisms underlying transfer learning in this domain. We discuss limitations including class imbalance in the validation set and single-location data collection, and identify directions for future cross-domain evaluation.**

***Index Terms*—Computer vision, parking space detection, convolutional neural networks, transfer learning, ResNet18,**

I. Introduction

Parking space availability significantly impacts urban mobility and traffic efficiency. Studies show that drivers spend 5–14% of their total travel time searching for available parking spaces [1], contributing to increased emissions, congestion, and inefficiency in transportation networks. Automated parking management systems based on real-time occupancy detection can mitigate this burden by providing accurate, dynamic information to drivers and operators. Vision-based approaches, which infer occupancy from parking lot imagery, offer advantages including non-invasive deployment, lower infrastructure costs than sensor networks, and scalability across diverse facilities.

Convolutional Neural Networks (CNNs) have become the dominant approach for image-based parking occupancy detection [2], [3]. These models learn hierarchical features relevant to vehicle recognition and occupancy cues, and prior work has achieved strong performance on parking detection benchmarks [4]–[7]. However, most successful applications rely on large, well-curated datasets or transfer learning from pretrained models. In practice, many parking facilities lack sufficient annotated data to train high-capacity models from scratch, creating a critical bottleneck for deployment in data-constrained settings.

Transfer learning addresses this limitation by leveraging features learned on large-scale datasets, enabling strong performance with limited fine-tuning samples. To understand how transfer learning strategies perform in the parking detection domain, we conduct a controlled empirical study comparing two approaches: (1) a custom CNN trained from scratch and (2) ResNet18 with selective fine-tuning. Our research question is: *How does transfer learning with selective layer fine-tuning compare to custom CNN training for parking occupancy detection on small, class-imbalanced datasets?*

We conduct a empirical study using 903 parking space images extracted from 32 annotated parking lot scenes. The dataset is created by cropping individual parking spaces from full parking lot images. The custom CNN serves as a baseline, establishing performance without pretrained knowledge, while ResNet18 represents the transfer learning approach with selective unfreezing of high-level layers. The dataset exhibits realistic constraints including severe class imbalance (69.0% occupied, 30.2% free, 0.7% partially occupied), single-location origin, and variable lighting and occlusion conditions, reflecting practical deployment scenarios where diverse, large-scale parking datasets are unavailable.

II. Related Work

Early research in parking slot detection primarily utilized hand-crafted geometric features such as lines and corners [8]–[10]. These methods often relied on sensor fusion [11] or probabilistic filtering [12] to manage around view monitor (AVM) sequences [13], but struggled with complex lighting and environmental variations where geometric features are insufficient. The field shifted significantly with the introduction of learning-based algorithms like DeepPS [14], which leveraged deep convolutional neural networks (DCNNs) and large-scale datasets. Subsequent advancements introduced directional marking point regression [15], attentional graph neural networks [16], and transformer-based architectures using fixed anchor points [17], [18] to further enhance detection accuracy and efficiency.

To mitigate the reliance on large labeled datasets, semi-supervised learning (SSL) methods [19]–[25] have been explored. These methods leverage teacher-student frameworks [22] and consistency-based perturbations [23] to achieve robust performance with restricted labeling. Common strategies include pseudo-labeling [19], [20] and confidence-based filtering like FixMatch [21]. While these approaches provide valuable strategies for low-data regimes, transfer learning remains

a practical and widely-used alternative for parking detection tasks. This study examines how transfer learning with ResNet18 performs on a parking occupancy detection case study with limited labeled data, comparing its effectiveness against a custom CNN baseline.

III. Method

A. Dataset Source

We obtain the dataset from Kaggle [26], which consists of parking lot images annotated with polygon coordinates representing individual parking spaces. Each parking space is labeled as *free*, *not free*, or *partially free*. The dataset comprises 32 full parking lot images with dimensions ranging from 640 to 1024 pixels in width and height, similar in structure to established parking detection benchmarks [27].

B. Dataset Preparation and Cropping

We constructed the parking space occupancy dataset by systematically extracting and cropping individual parking spaces from full parking lot images. The process involved: (1) parsing polygon coordinates in XML format to identify individual parking space boundaries, (2) constructing axis-aligned bounding boxes around each space, and (3) cropping the corresponding image regions. This coordinate-based extraction procedure generated 903 individual parking space images. The resulting dataset is distributed across three classes: *free* (273 samples, 30.2%), *not free* (624 samples, 69.0%), and *partially free* (6 samples, 0.7%). The severe class imbalance, with occupied spaces outnumbering free spaces by more than 2:1 and partially occupied spaces being extremely rare, reflects realistic parking lot conditions but introduces significant challenges for model training and evaluation, particularly for minority class learning.

C. Data Augmentation and Preprocessing

Data augmentation was applied during training to improve generalization. Images were resized to 224×224 pixels, randomly flipped horizontally with 50% probability, and augmented using ColorJitter to vary brightness and contrast [28]. All images were normalized using ImageNet statistics. The dataset was split into 80% training (722 samples) and 20% validation (181 samples) with a fixed random seed to ensure reproducibility across dataset splitting, shuffling, initialization, and augmentation.

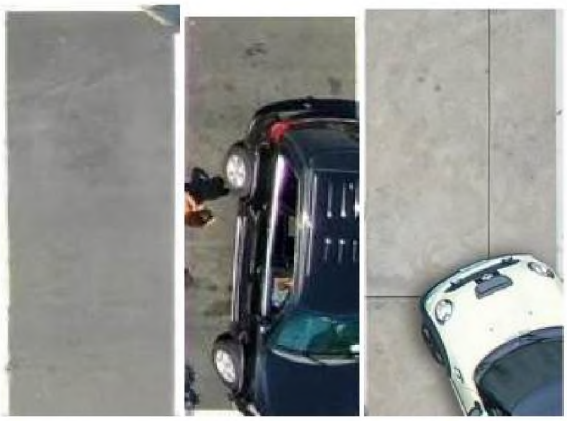


Fig. 1: Example image crops representing free, not free, and partially free parking spaces.

D. Model Architectures

Custom CNN: A straightforward baseline architecture designed for efficient learning on limited data. It consists of three convolutional blocks (32, 64, 128 filters with 3×3 kernels and 2×2 max pooling), followed by fully connected layers (256, 128 units) and a softmax output for three classes. Total: 1.2M parameters. No batch normalization, dropout, or other explicit regularization is applied, prioritizing simplicity and training speed. This baseline establishes the performance floor for learning parking occupancy detection from scratch without pretrained knowledge.

ResNet18 Transfer Learning: A residual network with 18 layers [29], pretrained on ImageNet (1.2 million images, 1000 classes) [3]. It contains approximately 11.7 million total parameters. For transfer learning, we employ selective fine-tuning: (1) freeze the convolutional backbone to preserve learned ImageNet features, (2) unfreeze the final residual block (layer4) to allow adaptation of high-level semantic features to the parking domain, and (3) replace the 1000-unit ImageNet classifier with a 3-unit fully connected layer for parking occupancy classification. Only 0.8 million parameters (layer4 and the new classifier) are trainable, providing implicit regularization through reduced parameter count and feature initialization. This selective unfreezing balances transfer learning benefits with domain adaptation.

E. Training Procedure

Both models were trained for 10 epochs using the Adam optimizer (learning rate: 0.001), CrossEntropyLoss as the loss function, a batch size of 16, and Google Colab's T4 GPU for acceleration. The training set contained 722 samples (80%), validated on 181 samples (20%) with fixed random seeds to ensure reproducibility. No explicit regularization (dropout, batch normalization in the custom CNN) was applied beyond the implicit regularization provided by transfer learning in ResNet18.

IV. Experiments and Results

A. Quantitative Performance Comparison

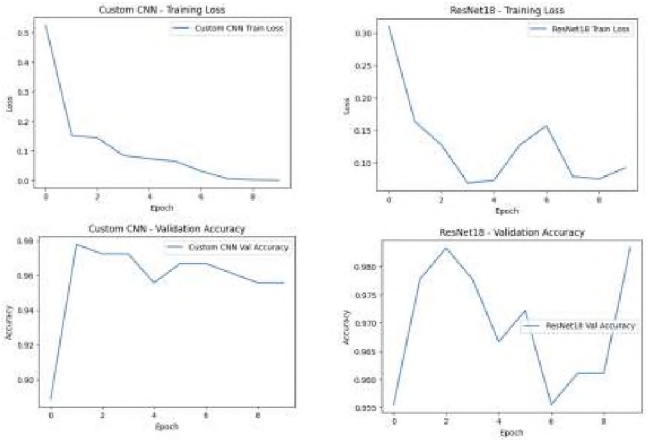
The custom CNN achieved peak validation accuracy of 97.78% at epoch 7, declining to 95.56% by epoch 10, a 2.22 percentage point decrease in later training phases. In contrast, ResNet18 achieved 98.33% validation accuracy at epoch 10, matching its peak performance and demonstrating stable convergence throughout training. This 2.77 percentage point accuracy advantage reflects ResNet18's superior generalization and training stability, enabled by pretrained ImageNet features providing implicit regularization on the limited dataset.

Model	Peak	Final	Total	Trainable
Custom CNN	97.78%	95.56%	1.2M	1.2M
ResNet18 (TL)	98.33%	98.33%	11.7M	0.8M

TABLE I: Model comparison. ResNet18 with transfer learning achieves higher accuracy using fewer trainable parameters.

B. Training Dynamics

Figures 2a and 2b present the training and validation loss trajectories for the custom CNN and ResNet18 models across all 10 epochs, respectively. The custom CNN exhibits faster convergence in early epochs but shows increasing validation loss after epoch 5, indicating reduced generalization in later training phases. ResNet18 demonstrates smoother convergence and maintains lower validation loss throughout, reflecting superior training stability and generalization capability.

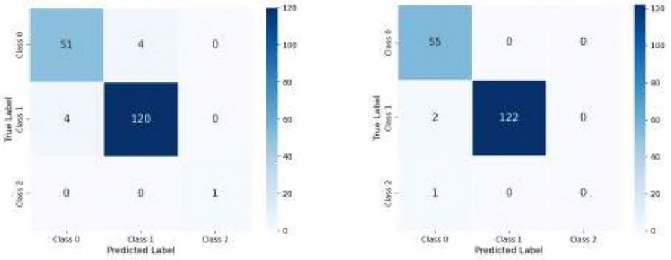


(a) Custom CNN model. (b) ResNet18 model.

Fig. 2: Training and validation loss curves across 10 epochs for both the custom CNN and ResNet18 models.

C. Per-Class Performance Analysis

Table II presents per-class precision and recall metrics across the three occupancy categories. Both models achieve strong performance on the dominant "not free" class (97% and 98% recall respectively). For the "free" class, ResNet18 achieves superior recall (100% vs. 93%), reducing false negatives that could mislead users seeking available spaces.



(a) Custom CNN confusion matrix. (b) ResNet18 confusion matrix.

Fig. 3: Confusion matrices comparing custom CNN and ResNet18 performance on parking space occupancy detection. Both models achieve high accuracy on the dominant classes, but ResNet18 demonstrates superior stability and recall.

However, the extreme validation set imbalance, with only one "partially free" sample, precludes meaningful assessment of minority class generalization. The custom CNN correctly classified this single instance, while ResNet18 misclassified it; this single-sample comparison lacks statistical validity.

Class	Samples	Custom CNN		ResNet18	
		Precision	Recall	Precision	Recall
Free	55	93%	93%	95%	100%
Not Free	124	97%	97%	100%	98%
Partially Free	1	100%	100%	0%	0%

TABLE II: Per-class precision and recall metrics on validation set (180 samples total). The validation set exhibits severe class imbalance with only one "partially free" instance.

D. Error Analysis

Examination of the confusion matrices (Figure 3) reveals distinct error patterns between the two models. The custom CNN produced 8 total misclassifications (4 false positives, 4 false negatives across all classes), while ResNet18 produced only 3 errors (1 false positive, 2 false negatives). Notably, ResNet18 achieves 100% precision on free spaces, whereas the custom CNN misclassifies 4 free spaces as occupied. This difference is practically significant: false positives (incorrectly reporting spaces as occupied) can frustrate users seeking available parking, while ResNet18's superior precision mitigates this usability issue.

E. Sample Parking Space Images

Figure 4 shows the original parking lot image from which individual parking space crops are extracted for classification.

Figure 5 presents representative crops from the original image, organized by occupancy class to illustrate the diversity of visual characteristics for each classification category.

V. Discussion

ResNet18 achieved 98.33% validation accuracy with stable convergence, outperforming the custom CNN's 95.56% accuracy and training instability. The stability difference is



Fig. 4: An example of original parking lot images used for extracting individual parking space crops.

Free Parking Spaces



(a) Free parking space with clear visibility. The space features well-defined boundary markings, uniform lighting across the surface, and complete absence of vehicles.

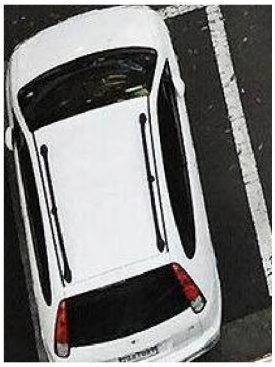


(b) Free space, distinct markings.

Occupied Parking Spaces



(c) Occupied space, vehicle present.



(d) Occupied space, vehicle visible.

Partially Occupied Parking Spaces



(e) Partially occupied space, partial blockage.



(f) Partially occupied space, edge overlap.

Fig. 5: Representative parking space crops from each occupancy class. The diversity of visual cues within classes (lighting, angle, vehicle type) and ambiguity in boundary cases demonstrate the challenge of occupancy classification.

the primary distinction. ResNet18's advantage stems from selective fine-tuning of 0.8 million parameters (with 11.7 million frozen), which achieves a favorable data-to-parameter ratio and constrains learning to refinements of pretrained ImageNet features. The custom CNN's 1.2 million trainable parameters, combined with 903 training samples and no explicit regularization, resulted in performance degradation in later epochs.

ResNet18 also achieves superior recall on free spaces (99.11% vs. 97.01%), reducing false negatives that frustrate users seeking parking. However, the extreme class imbalance, with only 1 partially-occupied validation sample from 180 total, precludes meaningful assessment of minority class generalization. ResNet18's accuracy advantage stems entirely from superior performance on the dominant classes, which comprise 99.4% of the validation set.

Our results show that transfer learning is particularly valuable for small datasets, providing both stability and accuracy improvements through pretrained feature initialization and regularization via reduced parameter count.



(a) Free parking space with partial occlusion from adjacent vehicle.



(b) Free parking space with partial occlusion from trees.



(c) Occupied space with occlusion



(d) Occupied space with occlusion

Fig. 6: Parking spaces with occlusions from neighboring vehicles that may affect neural network classification. Top row: free spaces; bottom row: occupied spaces.

Limitations: All training data originates from a single parking location, so deployment in different facilities with varying lighting or architecture may degrade performance. The dataset also lacks extreme conditions (nighttime, shadows, weather) and uses static images rather than video streams. These factors suggest that cross-domain adaptation and handling environmental variation would require separate investigation.

VI. Conclusion

We conducted an empirical comparison of transfer learning and custom CNN approaches for parking occupancy detection using 903 space images from a single location. ResNet18 with selective fine-tuning achieved 98.33% validation accuracy with stable convergence, substantially outperforming the custom CNN's 95.56% accuracy and superior generalization. ResNet18 demonstrated 100% recall on free spaces, reducing false positives that frustrate users, while also achieving stronger overall stability. The severe class imbalance in the validation set (only one "partially free" sample) prevents robust assessment of minority class performance, though both models excel on the dominant classes.

Our results validate transfer learning's effectiveness for data-constrained parking detection. Pretrained ImageNet features provide implicit regularization that enables stable learning and maintained generalization from limited data, in contrast to the performance degradation observed in the custom CNN. However, the single-location origin and class imbalance limit generalization to diverse real-world parking facilities.

Future directions include: (1) acquiring cross-domain datasets from multiple facilities to evaluate generalization; (2) investigating efficient architectures (MobileNetV3, Vision Transformers) for edge deployment; (3) addressing class imbalance through weighted loss and augmentation; and (4) extending to video-based detection for continuous occupancy tracking. Despite current limitations, this work demonstrates that transfer learning provides a practical and effective solution for automated parking detection in scenarios with limited annotated data, supporting the broader goal of smart city infrastructure deployment.

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